Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details:

Batch Number: 13

Start date: 02-02-2026

Name: Saniya Sheldarkar

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX AI-Driven Urban Mobility Optimization System

2. Project Objective

The primary objective of this project is to design and develop an intelligent urban mobility system that reduces traffic congestion and improves fleet efficiency. NeuroFleetX aims to optimize vehicle routing, predict maintenance needs, and enhance user experience through AI-driven recommendations. A key performance indicator (KPI) considered in this project is **a**verage travel time per trip, which reflects the system's effectiveness in traffic optimization.

3. Project description in detail

NeuroFleetX is an AI-powered platform designed to enhance urban mobility by integrating real-time data, machine learning, and intelligent decision-making systems.

The system includes multiple modules such as authentication, fleet tracking, route optimization, predictive maintenance, and customer booking. The backend is developed using Spring Boot, while the frontend uses React/Angular for interactive dashboards. AI functionalities are implemented using Python-based microservices, and data is managed using **MySQL.**

The platform uses simulated IoT data for vehicle telemetry, including location, speed, and fuel/battery levels. AI-based routing algorithms (such as Dijkstra combined with predictive models) help determine optimal routes considering traffic conditions. Predictive maintenance alerts are generated based on vehicle health parameters.

In real-world implementation, NeuroFleetX can be used in smart cities, ride-sharing systems, logistics, and EV fleet management. It helps reduce congestion, improve efficiency, and promote sustainable mobility solutions.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Project understanding, requirement analysis	Finalized project idea and modules
Week 2	UI design and authentication setup	Designed login/register screens
Week 3	Backend setup and database design	Implemented APIs and schema
Week 4	Fleet module development	Vehicle CRUD and telemetry simulation
Week 5	Route optimization integration	Map APIs and route logic implemented
Week 6	Predictive maintenance module	Health analytics dashboard created
Week 7	Booking system and recommendations	Customer UI and AI logic completed
Week 8	Testing, optimization, documentation	Final deployment and report prepared

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial planning and team alignment	Week 1
Prototype/First Draft	Basic UI and backend setup	Week 3
Mid-Term Review	Core modules completed	Week 5
Final Submission	Fully integrated system	Week 8
Presentation	Final demo and explanation	Week 8

5b. Project execution details

Milestone 1: 13-Feb-2026

In Milestone 1, I implemented the Authentication and Role Management module of the NeuroFleetX system using Spring Boot and React. Secure user registration and login were developed using JWT-based authentication. Backend APIs were created in AuthController.java and UserService.java, with database connectivity configured in application. properties using MySQL. Registered user data is stored in the users table and verified during login. The frontend includes login, registration, and role-based dashboard redirection for Admin, Manager, Driver, and Customer.

Milestone 2: 03-March-2026

A full fleet inventory system has been implemented, including vehicle CRUD operations where vehicles can be added, updated, and managed through the dashboard. Real-time status tracking has also been implemented for each vehicle, such as available, in use, maintenance, and offline. In addition, telemetry simulation has been integrated to generate dynamic data like speed, fuel level, battery percentage, and GPS location. All this information is displayed in a grid-based vehicle card layout with status indicators and progress bars.

Milestone 3: 12-Mar-2026

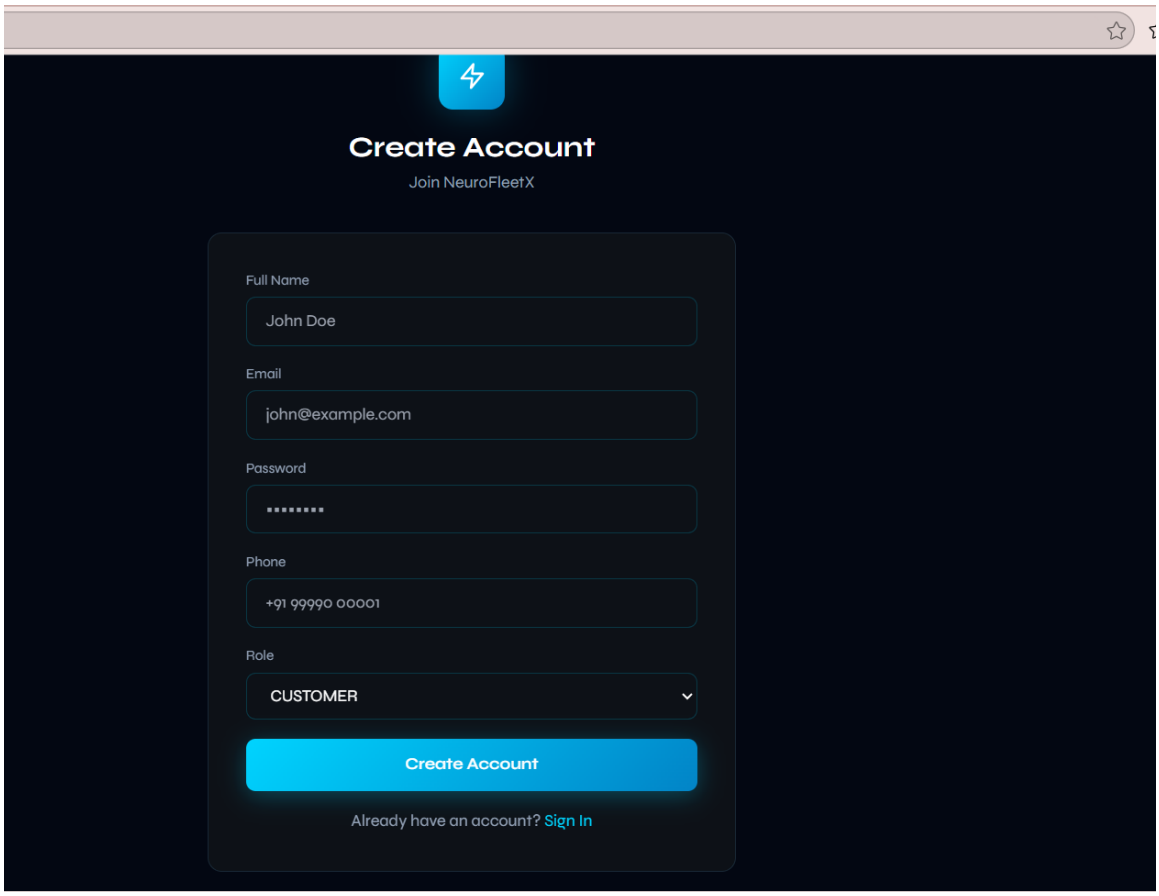
In Milestone 3 of the NeuroFleetX project, I began by studying how vehicles can be allocated efficiently based on their availability. After that, I explored shortest path concepts to understand how optimal routes are determined. I then worked on estimating travel distance and time to generate an approximate ETA using simple logic. Finally, I integrated map functionality to visually display the route between the starting point and destination. Through this milestone, I developed a better understanding of how route optimization plays a key role in improving fleet efficiency.

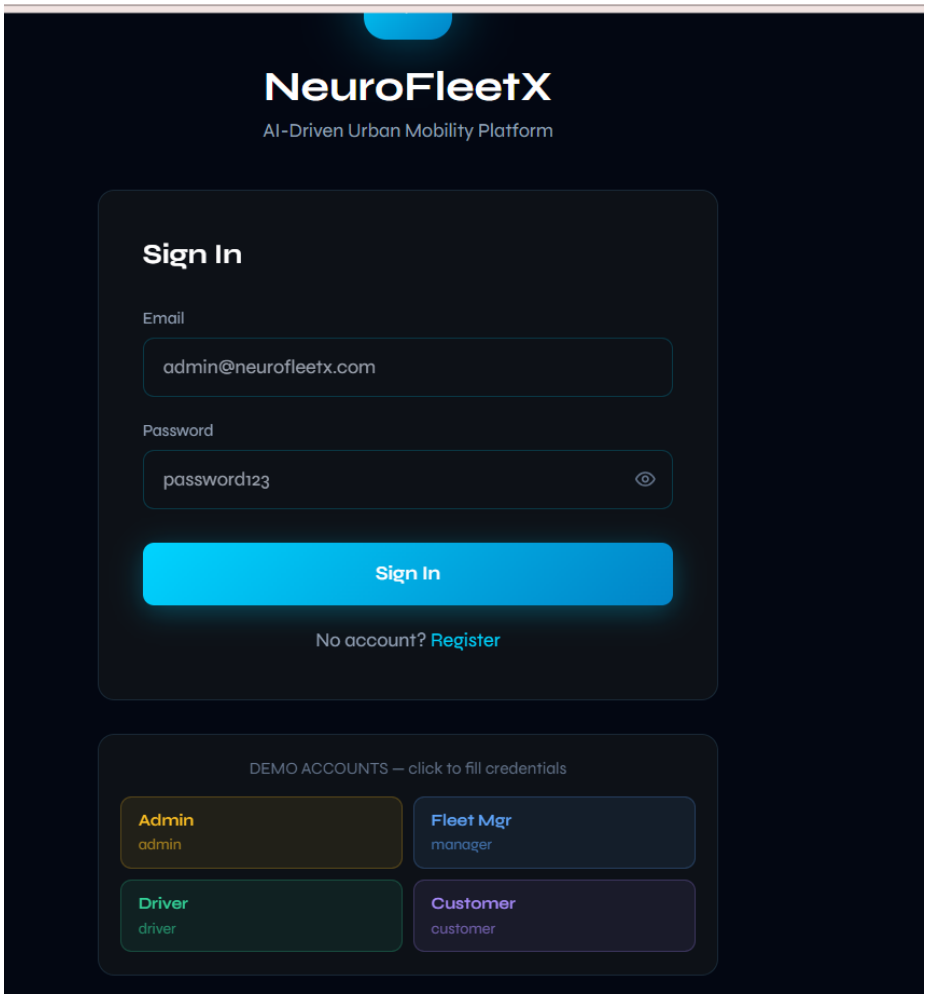
Milestone 4: 23-Mar-2026

In this Milestone 4, I have implemented Predictive Maintenance This module monitors vehicle health parameters such as engine, tyres, battery, and fuel. Alerts are generated when values go below threshold, helping prevent breakdowns. as well as implemented Booking & Recommendation System ,This module allows users to book vehicles. Based on input like passengers and distance, the system recommends the best vehicle. Booking status is tracked throughout the process.

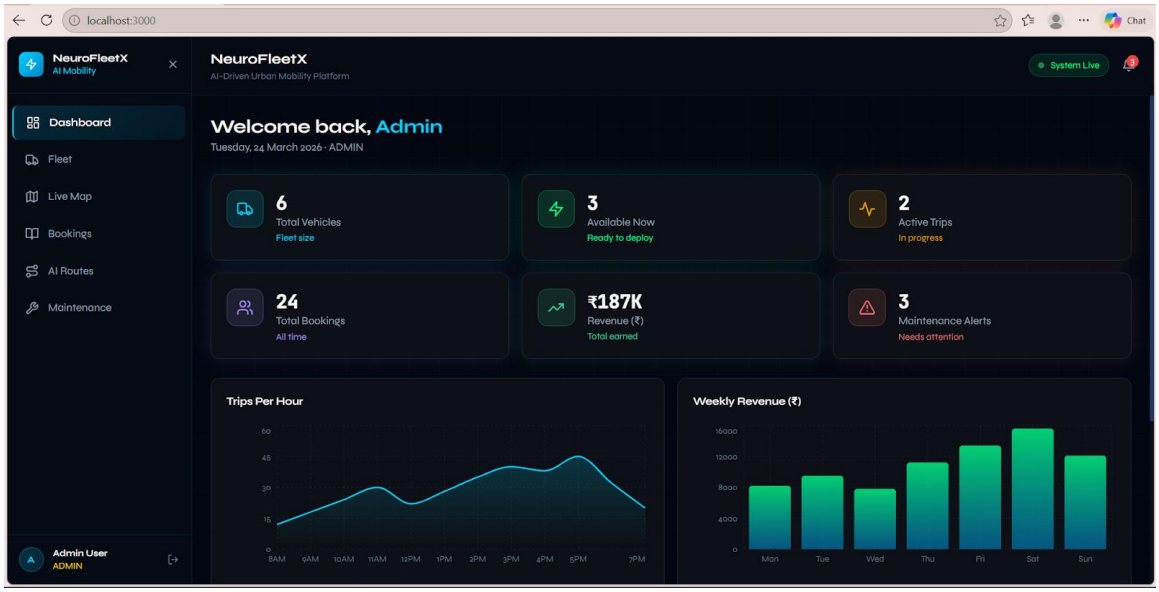
6. Snapshots / Screenshots

Register UI and Login UI



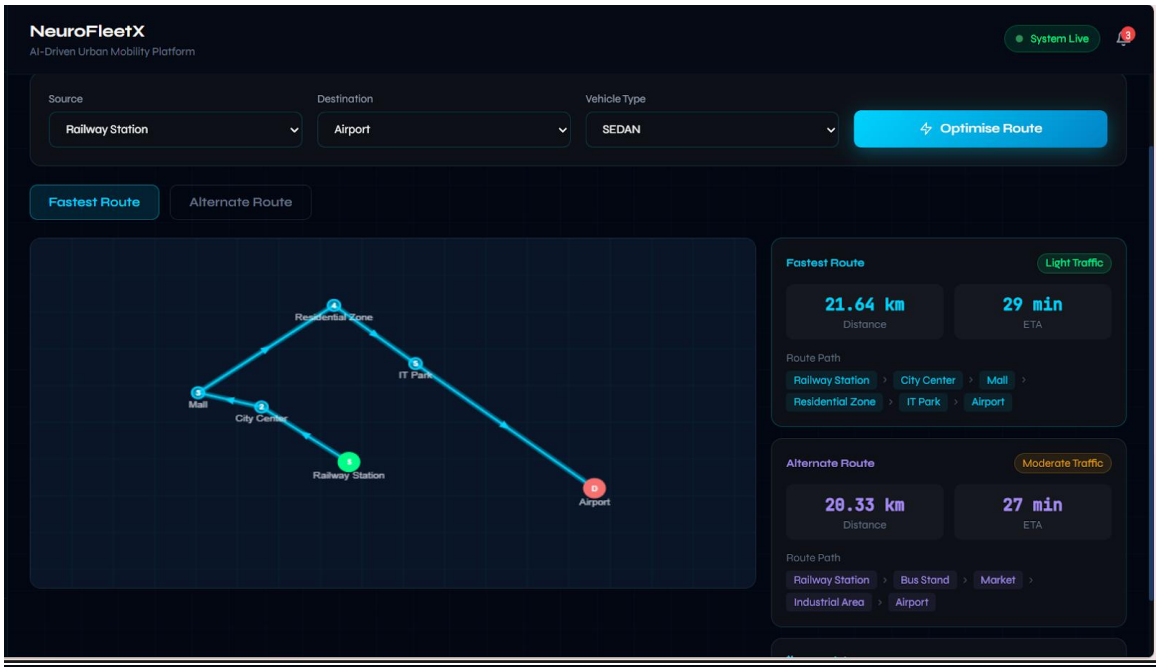

NeuroFleetX
AI-Driven Urban Mobility Platform
Sign In
Email
admin@neurofleetx.com
Password
password123
Sign In
No account? Register
DEMO ACCOUNTS — click to fill credentials
Admin
admin
Fleet Mgr
manager
Driver
driver
Customer
customer

Dashboard

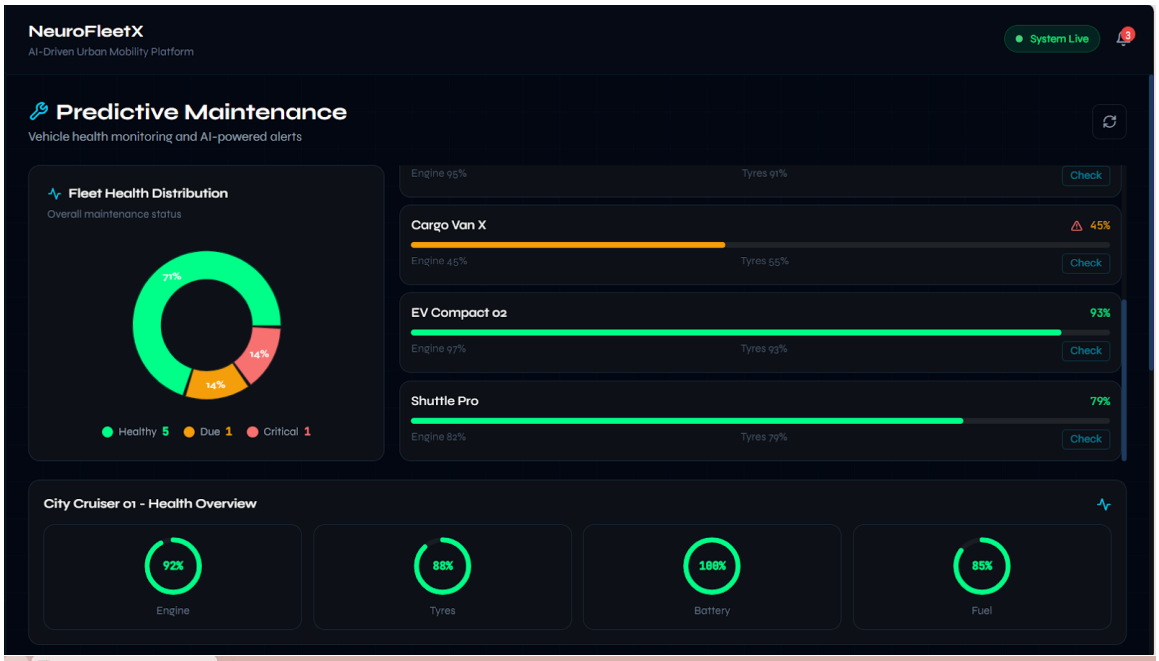


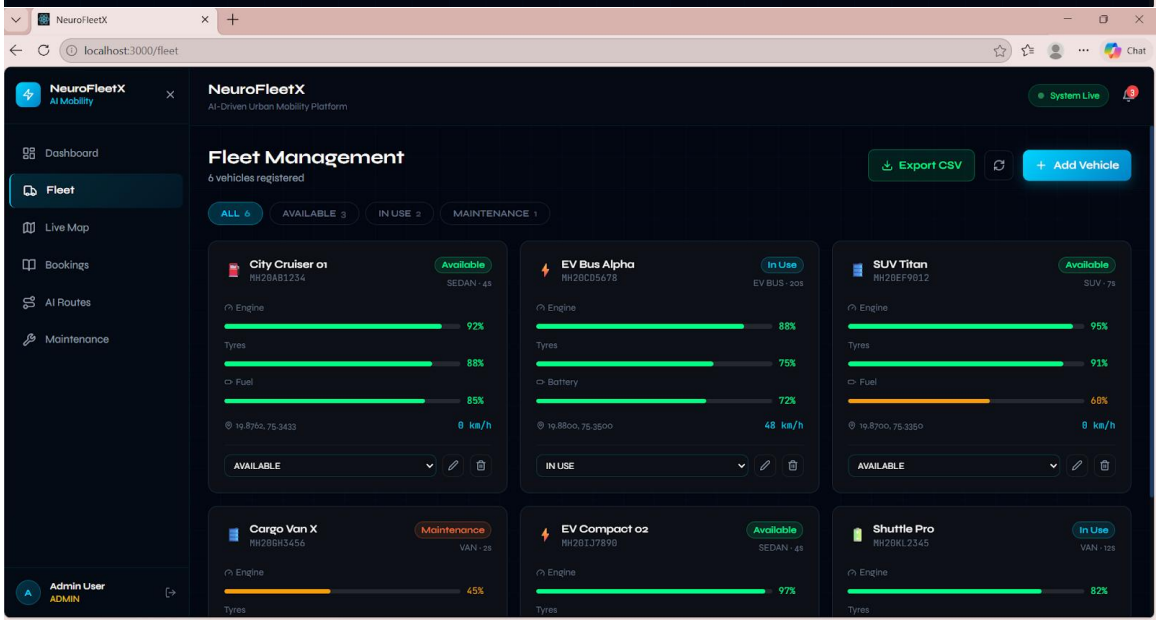
Fleet cards

Map with routes



Analytics charts





7. Challenges Faced

During the project, several challenges were encountered:

Integration complexity: Combining frontend, backend, and AI modules required careful coordination.

API handling: Working with map APIs and real-time simulation was initially difficult.

Data simulation: Creating realistic vehicle telemetry data required multiple iterations.

Team coordination: Managing tasks across multiple modules needed proper planning.

These challenges were resolved through team discussions, research, and iterative testing.

8. Learnings & Skills Acquired

This internship helped in gaining both technical and soft skills:

Technical Skills:

Spring Boot & REST APIs

React/Angular frontend development

Database design using MySQL

AI concepts and predictive analytics

API integration (Maps, WebSockets)

Soft Skills:

Team collaboration

Problem-solving

Time management

Project planning

9. Testimonials from team

Working on NeuroFleetX was a highly enriching experience. The project helped us understand real-world applications of AI in smart mobility. We learned how to work as a team, divide responsibilities, and deliver a complete product within deadlines. The internship improved our confidence and technical abilities significantly.

10. Conclusion

The NeuroFleetX project successfully demonstrates how AI can be used to optimize urban mobility systems. It provides solutions for reducing traffic congestion, improving fleet efficiency, and enhancing user experience.

This internship aligned well with our academic learning and provided practical exposure to real-world technologies. It has strengthened our interest in AI, full-stack development, and smart city solutions.

11. Acknowledgements

We would like to express our sincere gratitude to Infosys Springboard for providing this valuable internship opportunity. We also thank our mentor for their continuous guidance and support. Lastly, we appreciate our team members for their dedication and collaboration throughout the project.